**This assignment will not be graded and is only for practice.**

## Level 1

1) Choose the set of correct options. ***1 point***

- [ ] If $v_1$ and $v_2$ are in the kernel of a linear transformation $T : V \rightarrow W$, where $V$ and $W$ are two vector spaces, then $v_1 + v_2$ will also be in the kernel of $T$.
- [ ] Kernel of a linear transformation $T : V \rightarrow W$ is a vector subspace of $V$, where $V$ and $W$ are two vector spaces.
- [ ] If $w_1$ and $w_2$ are in the image of a linear transformation $T : V \rightarrow W$, where $V$ and $W$ are two vector spaces, then $w_1 + w_2$ will also be in the image of $T$.
- [ ] Image of a linear transformation $T : V \rightarrow W$ is a vector subspace of $W$, where $V$ and $W$ are two vector spaces.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $v_1$ and $v_2$ are in the kernel of a linear transformation $T : V \rightarrow W$, where $V$ and $W$ are two vector spaces, then $v_1 + v_2$ will also be in the kernel of $T$.
Kernel of a linear transformation $T : V \rightarrow W$ is a vector subspace of $V$, where $V$ and $W$ are two vector spaces.
If $w_1$ and $w_2$ are in the image of a linear transformation $T : V \rightarrow W$, where $V$ and $W$ are two vector spaces, then $w_1 + w_2$ will also be in the image of $T$.
Image of a linear transformation $T : V \rightarrow W$ is a vector subspace of $W$, where $V$ and $W$ are two vector spaces.

Consider the following linear transformation:

$T : \mathrm{R}^3 \rightarrow \mathrm{R}^2$

$T(x, y, z) = (2x + 3z, 4y + z)$

**Answer questions 2,3,4 and 5, using the information given above.**

2) Which of the following matrices corresponds to the given linear transformation $T$ with respect to the standard ordered basis for $\mathrm{R}^3$ and the standard ordered basis for $\mathrm{R}^2$ ? ***1 point***

- $\begin{bmatrix} 2 & 3 & 0 \\ 4 & 1 & 0 \end{bmatrix}$
- $\begin{bmatrix} 2 & 0 \\ 0 & 4 \\ 3 & 1 \end{bmatrix}$
- $\begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$
- $\begin{bmatrix} 2 & 4 \\ 3 & 1 \\ 0 & 0 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$

3) Which of the following represents a basis of the kernel of $T$? ***1 point***

- $\{(-\frac{3}{2}, 0, 1), (0, -\frac{1}{4}, 1)\}$
- $\{(-\frac{3}{2}, -\frac{1}{4}, 1)\}$
- $\{(-\frac{3}{2}, -\frac{1}{4}, 2)\}$
- $\{(2, 0, 3), (0, 4, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-\frac{3}{2}, -\frac{1}{4}, 1)\}$

4) What will be the dimension of the subspace $Im(T)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

***1 point***

5) Choose the correct option. ***1 point***

- ○ $T$ is an isomorphism.
- ○ $T$ is one to one but not onto.
- ○ $T$ is onto but not one to one.
- ○ $T$ is neither one to one, nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$ is onto but not one to one.

6) Choose the set of correct options. ***1 point***

- ☐ There exists a linear transformation $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$ whose range is contained in the circle $x^2 + y^2 = 1$.
- ☐ There exists a linear transformation $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$ whose range equals the line $x = y$.
- ☐ There exists a linear transformation $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$ whose range is contained in the set $S = \{(x, y) \mid x > 0\}$.
- ☐ There exists a linear transformation $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$ whose range equals the line $y = \pi x$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exists a linear transformation $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$ whose range equals the line $x = y$.
There exists a linear transformation $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$ whose range equals the line $y = \pi x$.

## Level 2

7) Which option represents the kernel and image of the following linear transformation? ***1 point***

$T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$
$T(x, y) = (x, 0)$

- ○ $ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(1, 0)\}$
- ○ $ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(0, 1)\}$
- ○ $ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(1, 0)\}$
- ○ $ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(1, 0)\}$

8) Which of the following linear transformations is an isomorphism between the vector spaces $V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in \mathrm{R}\} \subset \mathrm{R}^3$ and $W = \{(x, y, z) \mid x = y, \text{ and } x, y, z \in \mathrm{R}\} \subset \mathrm{R}^3$? ***1 point***

- ○ $T : V \rightarrow W$, such that $T(x, y, z) = (y, y, 0)$
- ○ $T : V \rightarrow W$, such that $T(x, y, z) = (y, y, z)$
- ○ $T : V \rightarrow W$, such that $T(x, y, z) = (x, y, z)$

- ○ There does not exist any isomorphism between $V$ and $W$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T : V \to W$, such that $T(x, y, z) = (y, y, z)$

9) Which of the following linear transformations is an isomorphism between the vector spaces $V = \{(x, y, z) \mid x = y - z = 0, \text{ and } x, y, z \in \mathrm{R}\} \subset \mathrm{R}^3$ and $W = \{(x, y, z) \mid x = y = 0, \text{ and } x, y, z \in \mathrm{R}\} \subset \mathrm{R}^3$? ***1 point***

- ○ $T : V \to W$, such that $T(x, y, z) = (x, 0, y)$.
- ○ $T : V \to W$, such that $T(x, y, z) = (0, y, z)$.
- ○ $T : V \to W$, such that $T(x, y, z) = (0, 0, x)$.
- ○ There does not exist any isomorphism between $V$ and $W$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T : V \to W$, such that $T(x, y, z) = (x, 0, y)$.

10) Consider the linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^3$ such that $T(x, y, z) = (x - y + z, 2x + y - z, -x - 2y + 2z)$. Which of the following options are true about $T$? ***1 point***

- ☐ Range of $T$ is $\{(x, y, z) \in \mathrm{R}^3 \mid x = y + z\}$.
- ☐ Range of $T$ is $\{(x, y, z) \in \mathrm{R}^3 \mid x = 0, y + z = 0\}$.
- ☐ Rank of $T$ is $2$
- ☐ Rank of $T$ is $1$

No, the answer is incorrect.

Score: 0

## Accepted Answers:

Range of $T$ is $\{(x, y, z) \in \mathrm{R}^3 \mid x = y + z\}$.
Rank of $T$ is $2$

Your score is: 0/10